Domain Robustness of Multilingual NLP Models Across Urdu and Roman Urdu Scripts

MUHAMMAD ABDULLAH HAROON and MARYAM BASHIR, FAST School of Computing, National University of Computer and Emerging Sciences (FAST-NUCES), Pakistan

Urdu classifiers are almost always trained and tested on a single corpus, so little is known about how they behave when the data source changes. We study this question with a benchmark of 26 Urdu classification datasets (15 from public corpora and 11 generated with an LLM; 188,309 labelled examples) arranged into 7 domain-transfer matrices over four task types (sentiment and polarity, abusive language, fake news, and question–answer pair validation) in two scripts, Nastaliq and Roman Urdu. Because the benchmark combines public corpora with generated text, its construction is itself a contribution that applies to other low-resource languages: explicit inclusion criteria that exclude duplicated domains, domains whose label can be read off the origin of an item, and domains too small or too imbalanced to evaluate; a single-phrase shortcut audit; splits that keep near-duplicate items and questions out of the test data; and measurements that separate generated from human-written text. Following Calderon et al. [13], we evaluate two fine-tuned encoders (XLM-R, mBERT) and three few-shot instruction-tuned LLMs (Llama-3.1-8B, Qwen2.5-7B, Mistral-7B) on all 80 source-to-target shifts, with three seeds or demonstration draws, three baselines and paired bootstrap tests. Fine-tuned encoders lose 13.5–47.6 macro-F1 points when moved to another domain, and in 83–100% of shifts a model trained on the target does better, so these losses are robustness failures rather than moves to harder domains. Few-shot LLMs lose far less; their cross-domain F1 is significantly higher in 30 of 42 comparisons after Holm correction and significantly lower in 4, all in QA validation. The largest failures of both families arise where two domains attach the same label to different kinds of text. Across the two fake-news corpora available, neither family transfers usefully. Data, code and prompts are available at https://github.com/AbdullahPatti/Urdu-Multi-Domain-Script-Research.

CCS Concepts: • **Computing methodologies** → **Natural language processing**; *Language resources.*

Additional Key Words and Phrases: Domain robustness, Urdu, Roman Urdu, low-resource NLP, benchmark construction, shortcut learning, in-context learning

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1 INTRODUCTION

Urdu is spoken by more than 230 million people, yet its NLP resources are fragmented and small compared with those of high-resource languages [15]. Most Urdu classifiers are trained and tested on a random split of one corpus. Such an evaluation measures how well a model fits one distribution; it does not tell a practitioner whether the model will keep working when the data source changes (a new platform, topic or publisher) while the language, script and task stay the same.

Authors' address: Muhammad Abdullah Haroon, l230734@lhr.nu.edu.pk; Maryam Bashir, maryam.bashir@lhr.nu.edu.pk, FAST School of Computing, National University of Computer and Emerging Sciences (FAST-NUCES), Lahore, Pakistan.

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This property, *domain robustness*, has been studied extensively for English [8, 28, 46]. Most recently, Calderon et al. [13] evaluated fine-tuned models and few-shot LLMs over more than 14,000 domain shifts in seven English tasks and showed that the usual measure, the drop from in-domain to cross-domain performance, overstates robustness problems unless it is read together with a second measure that asks whether the target domain is simply harder.

No comparable study exists for Urdu, and Urdu adds a complication English lacks: it is written in two scripts that occupy different niches. Nastaliq (Perso-Arabic, right-to-left) dominates edited text such as news, whereas Roman Urdu (Latin, left-to-right) dominates informal online text, has no standard spelling (*bohat*, *bht* and *bohot* all mean "very") and mixes freely with English [5, 48]. Table 1 shows one example of each task type. Because the two scripts come with different sources, registers and available tasks, we treat them as separate evaluation tracks.

Building a multi-domain benchmark from public Urdu corpora raises a second problem, which is the focus of our data work. Public corpora overlap, some are published in forms in which the label follows the source of an item, and domains expanded with LLM-generated text contain templated near-duplicates. Each of these makes cross-domain scores look better or worse than they are. We therefore select domains by stated criteria and split them so that such artefacts cannot reach the test data.

We ask four research questions:

RQ1 How much do multilingual models lose when an Urdu classifier is moved to another domain of the same task, and is the loss a robustness failure or a move to a harder domain?

RQ2 Does the English finding that few-shot LLMs transfer better than fine-tuned encoders hold for Urdu and Roman Urdu?

RQ3 What causes the largest failures?

RQ4 Do the answers change when LLM-generated domains are excluded?

Contributions.

(1) **A multi-domain Urdu benchmark** of 26 labelled datasets in two scripts (188,309 examples), each with a documented origin, licence and class distribution, and splits that are free of exact and near-duplicate train–test overlap.

(2) **A set of inclusion criteria and audits** for building such benchmarks from public and generated corpora: detection of duplicated domains, a single-phrase shortcut audit that finds labels predictable from the origin of an item, question-grouped splitting for pair tasks, cluster-aware splitting for generated text, and measured signals that separate generated from human-written domains. None of these steps is specific to Urdu. We report how many candidate domains each criterion removes.

(3) **A domain-robustness evaluation following Calderon et al. [13]**, with in-domain, cross-domain and target in-domain scores, Source and Target Drop per shift, majority, random and length-only baselines, two degeneracy guards, three seeds, and Holm-corrected paired bootstrap tests.

(4) **A replication** of the fine-tuned versus few-shot comparison in a new language and script setting, with an error analysis that traces the largest failures to differences in what a label means from one domain to another.

2 BACKGROUND AND RELATED WORK

2.1 Terminology and problem formulation

A *domain* is a data source for a fixed language, script and task [13]: two domains share the label set and differ only in where their text comes from (platform, publisher, topic or collection effort).

Table 1. One example from each task type, in the script(s) in which the benchmark covers it. Items are quoted exactly from the test or training split of the named domain (IDs as in Table 2); the English glosses are ours. Nastaliq is written right to left in a cursive, largely standardised spelling. Roman Urdu is written left to right with phonetic, non-standard spelling (*bhut* for *bohat* "very", *neext*) and frequent English words (*project*, *bro*).

Matrix	Domain	Item	English gloss	Label
Nastaliq sentiment	SA-E	کسانوں کی محنت رنگ لائی اور حالیہ دنوں میں فصل کی پیداوار ریکارڈ سطح پر پہنچ گئی	The farmers' hard work paid off, and crop output has recently reached a record level.	Positive
Roman sentiment	SA-B	*theek he . bohat acha ha*	It's fine. Very good.	Positive
Roman 3-class polarity	PR-D	*mashallah bhut achy shyr hai*	Mashallah, a very good couplet.	Positive
Roman 3-class polarity	PR-D	*main bhi tumhe nahi usko bol raha hu*	I'm not saying it to you either, I'm saying it to him.	Neutral
Nastaliq hate speech	HS-A	سندھ کا پیسہ کہاں گیا؟ جواب دو چوروں نا اہلوں	Where did Sindh's money go? Answer, you thieves, you incompetents.	Hate/offensive
Roman cyber abuse	CA-A	*good luck for neext project bro*	Good luck for the next project, bro.	Normal
Nastaliq fake news	FN-B	لاپتہ پاکستانی کوہپیما زندہ مل گیا ہے۔	The missing Pakistani mountaineer has been found alive.	Fake
Nastaliq QA validation	QA-D	سوال: حقیقت میں انٹرنیٹ کیا ہے؟ جواب: دنیا بھر کے کمپیوٹرز کا نیٹ ورک	Q: What actually is the internet? A: A network of computers around the world.	Valid pair

For a task with domains $\mathcal{D} = \{D_1, \ldots, D_n\}$, let $F(i \to j)$ denote the macro-F1 on the test split of D_j of a model trained (or prompted) with data from D_i. A *shift* is an ordered pair (i, j) with $i \neq j$; a task with n domains has $n(n-1)$ shifts. The quantities computed from F are defined in Section 4.2.

2.2 Domain robustness and distribution shift

Models degrade under distribution shift [6, 35], and in NLP such shifts arise from topic, genre, platform and annotation conventions [8, 39]. Adaptation methods range from feature correspondence [9] and adversarial training [21] to continued pretraining [24]; Ramponi and Plank [46] survey them. Pretrained transformers are more robust than earlier architectures but still degrade [28]. Calderon et al. [13] show that the conventional Source Drop conflates genuine robustness failures with shifts to intrinsically harder targets, introduce the Target Drop to separate the two, and find that few-shot LLMs often transfer better than fine-tuned models while fine-tuned models are stronger in-domain. We adopt their framework.

2.3 Multilingual encoders and few-shot LLMs

mBERT [17] and XLM-R [14] transfer across languages [44, 52], and region-specific encoders such as IndicBERT [30] and MuRIL [33] cover Urdu and transliterated text. Instruction-tuned LLMs classify from in-context demonstrations without weight updates [12]; their accuracy is sensitive to demonstration choice, order and label calibration [37, 40, 54], which is why we use several demonstration draws.

2.4 Urdu and Roman Urdu NLP

Urdu resources exist for sentiment [31, 32], abusive and hateful language [2], and fake news [3, 26]; Roman Urdu resources cover sentiment [38, 50] and hate speech [47]. These corpora differ in annotation guidelines and are almost always evaluated on a random split of one source. Cross-dataset transfer for Urdu fake news alone is studied in our earlier preprint [25], which fine-tunes XLM-R across fake-news corpora and reports that article length separates fake from real items in some of them. The present study differs in scope and design: fake news is one of seven transfer matrices over four task types and two scripts, the fake-news corpora are admitted only if they pass the inclusion criteria of Section 3.3, few-shot LLMs are compared with two encoders, every shift is decomposed into Source and Target Drop, and the length analysis of that preprint becomes a length-only baseline that is applied to every matrix.

2.5 Leakage, shortcuts and synthetic data

Train–test overlap inflates reported scores and is a known threat to NLP evaluation [10, 20, 36]; near-duplicates are commonly detected with MinHash [11]. Classifiers also exploit superficial cues that correlate with the label in a given dataset, which makes in-domain scores optimistic and cross-domain scores hard to interpret. Machine-generated data is increasingly used to build classification corpora, including for hate speech [27], and it brings its own regularities. Section 3 shows how each of these arises when public and generated Urdu corpora are combined.

3 BENCHMARK CONSTRUCTION

3.1 Task types and domains

We collected 33 candidate datasets for four task types in the two scripts; the 26 that meet the inclusion criteria of Section 3.3 form the benchmark (Table 2, Figure 1). Domains are grouped into 7 *transfer matrices*. All domains in a matrix share one label space, so every source→target pair inside a matrix is a valid shift.

Sentiment and polarity. Nastaliq Sentiment (binary) has three domains: SA-A, tweets from the Urdu Sentiment Corpus [32]; SA-B, English IMDB movie reviews machine-translated into Urdu [42], whose content is almost entirely about Hollywood films; and SA-E, generated sentences on socio-economic and agricultural themes. *Roman Sentiment* (binary) has three: SA-A, mixed public discourse from a public Hugging Face release [4]; SA-B, the RUSAD reviews spanning sports, software, food, drama and politics [38]; and SA-C, YouTube comments on entertainment content. *Roman 3-class polarity* (negative/neutral/positive) combines two collections that share a label space. The first contains reviews and comments: PR-A, Daraz e-commerce reviews [53], which are code-mixed and often entirely English; PR-B, restaurant reviews; and PR-D, Roman Urdu reviews and comments from seven online sources [49], mostly on entertainment, politics and religion, whose five classes we merge into three. The second contains generated social-media opinions: TW-B on consumer electronics and gaming, TW-C on public and civic services, and TW-D on mixed social commentary. Domain IDs follow the candidate pool, so gaps in the lettering mark excluded candidates.

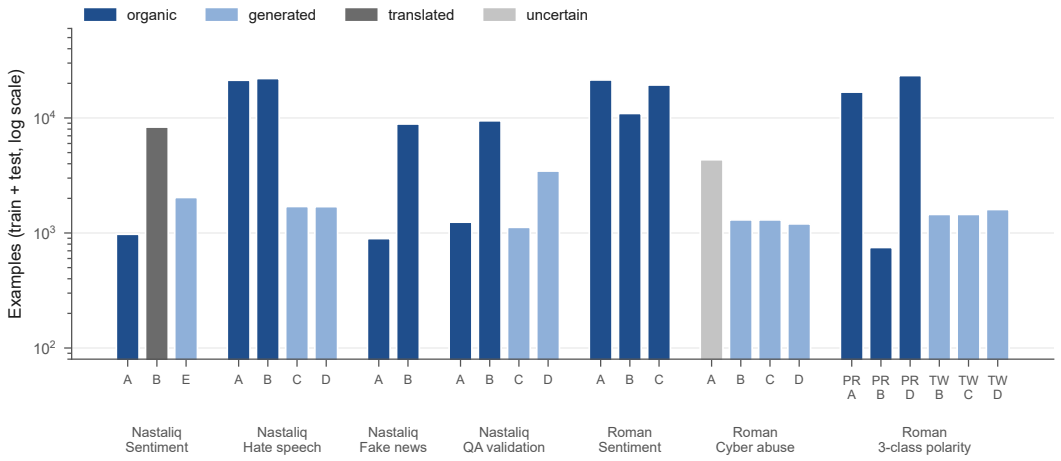


Fig. 1. Size of every domain (training and test items, log scale), grouped by transfer matrix and coloured by data origin. Exact counts and class distributions are in Table 2.

Abusive language. Nastaliq Hate Speech (normal vs. hate/offensive) has four domains: HS-A, the tweets of the ISE-Hate corpus [2]; HS-B, an independently annotated corpus of Urdu hate-speech tweets [1]; and HS-C and HS-D, generated text on politics, sports, health and family, and on sports, labour, arts and education. *Roman Cyber Abuse* (normal vs. abusive) has four: CA-A, YouTube comments; CA-B, generated comments aimed at content creators; CA-C, generated social-media abuse with diverse regional vocabulary; and CA-D, generated video comments.

Fake news. Nastaliq Fake News (fake vs. real) has two domains: FN-A, the "Bend the Truth" corpus, in which journalists wrote fabricated counterparts of business, health, showbiz, sports and technology news [3]; and FN-B, Ax-to-Grind Urdu, news and circulating claims from Pakistani and Indian outlets across fifteen topics [26].

QA pair validation. Nastaliq QA is a binary sequence-pair task, not generative QA. The input is a question and a candidate answer, labelled *valid* if the answer belongs to the question and *invalid* if it is a distractor drawn from another question of the same domain. QA-A (questions on Hadith texts) and QA-B (general-knowledge trivia) come from a public collection of Urdu datasets [41]; QA-C (long explanatory answers) and QA-D (short factual answers) are generated. Encoders read the question and answer as a sentence pair, LLMs answer with a label word, and the metric is macro-F1 as in every other matrix.

Appendix Table 21 gives the source, URL and licence of every domain. For each cited source we confirmed the origin by matching our files against the public release: our raw HS-B file is byte-identical to the public upload, and 91% of SA-B items match the public translated corpus once punctuation and letter variants are ignored. Three organic domains (Roman SA-C, CA-A and PR-B) came from public uploads whose original authors we could not identify; they are marked *untraced*.

3.2 Generated domains

Most task types have too few public Urdu corpora to form three or more domains. For the missing domains the first author generated data with xAI Grok. Each prompt described the target domain in detail (topic, platform, register and label definitions) and included examples from existing public

Table 2. The benchmark. C: number of classes; test distribution in label order (binary: negative/normal/fake/invalid first; 3-class: negative/neutral/positive). Origin: *organic* (public corpus of native text), *translated* (machine-translated from English), *generated* (LLM-generated, Section 3.2) or *uncertain*. Near-dup. %: share of test items of a random split that near-duplicated a training item (character 5-gram Jaccard $\geq$ 0.8). Split: *random* = stratified random split with leaking items removed; *cluster* = split so that near-duplicate clusters do not straddle train and test (Section 3.4).

Script	Transfer matrix	ID	Domain	C	N_{train}	N_{test}	Test dist.	Origin	Near-dup. %	Split
Nastaliq	Sentiment	SA-A	Twitter Social Media	2	781	194	99/95	organic	1.0	original
		SA-B	Movie Reviews	2	6,695	1,638	808/830	translated	0.1	original
		SA-E	Socioeconomic Agricultural	2	1,629	407	203/204	generated	43.0	cluster
	Hate Speech	HS-A	Social Media Offensive	2	17,152	4,133	2452/1681	organic	1.8	original
		HS-B	Social Media Hate Speech	2	18,486	3,540	1304/2236	organic	7.4	original
		HS-C	Politics Sports Health Family	2	1,360	340	120/220	generated	27.1	cluster
		HS-D	Sports Labor Arts Education	2	1,356	339	169/170	generated	38.6	cluster
	Fake News	FN-A	Multi Topic News	2	634	260	112/148	organic	0.0	original
		FN-B	Ax to Grind	2	7,217	1,645	789/856	organic	1.2	original
	QA Validation	QA-A	Religious Hadith	2	992	250	125/125	organic	0.0	original
		QA-B	General Knowledge	2	7,564	1,889	946/943	organic	0.2	original
		QA-C	Long Form QnA	2	896	224	112/112	generated	0.0	original
		QA-D	Short Form QnA	2	2,764	691	348/343	generated	0.7	original
Roman	Sentiment (binary)	SA-A	Mixed Public Discourse	2	19,262	2,140	1001/1139	organic	0.4	original
		SA-B	Social Media Politics Drama	2	8,781	2,163	1049/1114	organic	1.3	original
		SA-C	YouTube Entertainment	2	15,587	3,759	1630/2129	organic	0.7	original
	Cyber Abuse	CA-A	YouTube Comments	2	3,469	866	418/448	uncertain	13.2	cluster
		CA-B	Content Creator Comments	2	1,040	260	116/144	generated	63.5	cluster
		CA-C	Social Media Diverse Vocab	2	1,040	260	120/140	generated	59.6	cluster
		CA-D	General Online Video	2	960	240	110/130	generated	51.2	cluster
	3-class Polarity	PR-A	Daraz Ecommerce	3	13,489	3,294	850/487/1957	organic	1.4	original
		PR-B	Restaurant Food	3	600	148	8/23/117	organic	1.3	original
		PR-D	General Social Comments	3	19,349	4,026	1380/1158/1488	organic	0.3	original
		TW-B	Consumer Electronics Gaming	3	1,160	290	98/96/96	generated	90.7	cluster
		TW-C	Public Civic Services	3	1,160	290	98/96/96	generated	88.6	cluster
		TW-D	Mixed Social Commentary	3	1,280	320	110/103/107	generated	90.6	cluster
Total (26 domains)					154,703	33,606				

datasets of the same task, and asked for new labelled items in that domain. Claude and ChatGPT were tried first and discarded because their Urdu and Roman Urdu output was judged less natural. The first author, a native Urdu speaker, read all generated items and removed or corrected unnatural or mislabelled ones. Neither the prompts nor the number of removed items were recorded. None of the three evaluated LLMs was used for generation. To check these labels independently, a second native speaker of Urdu and Roman Urdu who took no part in generating or checking the data, Abdul Rehman Antall (a computer science student at FAST-NUCES with prior research experience), labelled a stratified sample of 330 generated test items (30 per generated domain) without seeing the original labels; he left three items blank. On the 327 labelled items his labels agree with the original ones on 95.7% (Cohen's $\kappa = 0.95$, pooled over domains; per domain 0.80 to 1.00; Appendix Table 19). In 12 of the 14 disagreements he chose the milder label (normal for an item labelled hate or abusive, neutral for one labelled negative), which agrees with our observation that some generated items are labelled more harshly than their wording supports. Naturalness judgements were not collected, so whether the items read as human-written rests on the measured signals of Appendix A.

Rather than rely on a recollection of how much of each domain was generated, we measured generation signals directly (Appendix Table 18): the share of items with a near-duplicate variant inside the domain, lexical diversity, and the variability of item length. The two groups separate cleanly on length: every generated domain has a coefficient of variation of item length of at most

0.22, and every organic domain at least 0.30; generated domains also reach template rates of up to 93%. The signals are uniform across each generated domain, so we treat all 11 such domains (18,306 examples, 9.7% of the benchmark) as generated in full. The share is concentrated: generated domains are one of three in Nastaliq sentiment, two of four in hate speech and QA, three of four in Roman cyber abuse and three of six in 3-class polarity, and none in fake news or Roman binary sentiment. Roman CA-A is recorded as organic but shows augmentation-like noise (inserted emoticons, repeated punctuation, misspellings) and a 13% near-duplicate rate, so we label it *uncertain*.

3.3 Inclusion criteria

A candidate domain enters the benchmark only if it meets the first three criteria below. Table 3 lists the 7 candidates that fail them. The fourth criterion was added after the experiments had been run, and its single failing domain is handled as described there.

(1) **Distinct source.** A domain must not repeat the text of another domain of the same matrix. Three candidates did: an ISE-Hate Level-1 file [2] that annotates the same tweet pool as HS-A, a political-tweet file identical to Nastaliq SA-A, and a second copy of the RUSAD corpus. Every shift between such a pair is effectively in-domain. Where only part of a domain overlapped another (e.g. 84% of Nastaliq SA-A tweets also occurred in SA-B), the shared texts were removed from the larger domain.

(2) **Label not predictable from origin.** A domain is excluded when its label can be read off where an item came from. Two candidates were distributed with each class taken from a different source: in one Roman opinion candidate every positive and negative item is a RUSAD review while every neutral item comes elsewhere, and in one Nastaliq fake-news candidate the fake items are largely fact-check verdicts and the real items ordinary news. To find less obvious cases we ran a *single-phrase shortcut audit* on every domain: for each word n-gram ($n \leq 3$) we compute the difference between its document frequency in one class and in the others, and inspect the top phrases (Appendix Table 20). In one further fake-news candidate the dateline token "92" (from "92 News") occurs in 89.6% of real items and 3.5% of fake items, so a model could classify it by recognising the outlet. All three were excluded. In the included domains the highest-scoring phrases are content words, such as *nahi* ("not") for negative opinions, with one exception discussed in Section 8.

(3) **Minimum size.** A domain needs at least 100 test items. One generated sentiment candidate had 33.

(4) **Minimum size per class.** Every class needs at least 30 test items. The size rule above checks only the total, and one included domain passed it with a nearly empty class: Roman PR-B has 148 test items, of which 8 are negative and 23 neutral, and its own XLM-R model scores below the random baseline of its matrix (Section 7.1). PR-B is the only included domain that fails this rule; the next smallest class in the benchmark has 95 test items. Because PR-B was part of the evaluated 3-class matrix, we keep it in the released benchmark and the main tables so that the reported runs stay reproducible, and we report every aggregate of that matrix both with and without it (Table 10).

3.4 Leakage-aware splits

Domains distributed with an official split (Nastaliq SA-B, FN-A and Roman SA-A) keep it. The other non-QA domains were split 80/20 at random, stratified by label (seed 42), after removing test items whose text also occurred in training. Exact de-duplication is not enough for generated text. MinHash near-duplicate detection [11] (character 5-grams, verified Jaccard ≥ 0.8) showed that in

Table 3. Candidate domains excluded by the inclusion criteria of Section 3.3.

Script	Collection	Domain	Reason
Nastaliq	Hate Speech	Inter Faith Sectarian Ethnic	same tweets and labels as Hate Speech A (ISE Level-1 re-uses the 22K Offensive pool)
Nastaliq	Sentiment Analysis	Governance Political	same texts and labels as Sentiment A
Roman	Sentiment Analysis	Utilities Apps Courses	same 11,000-review corpus as Roman Sentiment B
Roman	Twitter or Social Media Opinions	Twitter Political Social	positive/negative items are the RUSAD corpus (= Roman Sentiment B); all neutral items come from a different source, so the label is predictable from the source
Nastaliq	Fake News Detection	General Pakistani News	fake and real items come from two separate files; fake items are largely fact-check verdicts, so the label is predictable from the source
Nastaliq	Fake News Detection	Fact Checking Platform	real items carry an outlet dateline ("92 News", 90% of real vs 3.5% of fake items), so the label is predictable from the source
Nastaliq	Sentiment Analysis	Civic Social Topics	generated; too small to evaluate (162 items, 33 in test; minimum is 100 test items)

the generated non-QA domains 27–91% of test items of a random split near-duplicated a training item, against at most 7% in any organic domain except CA-A (13%). The duplicates are the same sentence with a filler added or swapped (*"…hain bhai"* vs. *"…hain bhai ok"*), a form of train–test leakage [20]. For the 10 domains in which more than 10% of test items leaked, near-duplicates were grouped into clusters and the domain was split 80/20 so that whole clusters fall on one side (stratified group split, seed 42); elsewhere the leaking test items were removed.

Pair tasks need a further step. A random split of question–answer pairs places the same question in training and test: in our QA domains this happened for 28–84% of test questions. QA domains are therefore split by *question group*: rows are linked when their questions are identical or near-duplicates, or when they share a distinctive answer of four or more words, and whole groups are assigned to training or test (80/20, seed 42). Pairs are then built inside each split, every question with its own answer (valid) and with an answer from a different group (invalid), so classes are balanced and no distractor answers a duplicate of the question.

After these steps no test item in any domain has an exact or near-duplicate in its training split. Fine-tuning holds out a further 10% of the source training split, stratified by label, for early stopping (Section 4.4). The build is deterministic: rebuilding under different Python hash seeds produces byte-identical files.

3.5 Licensing and release

Appendix Table 21 lists the licence of every source. Seven sources are released under MIT, CC BY 4.0 or ODbL 1.0. Four published corpora (the Urdu Sentiment Corpus, ISE-Hate, Bend the Truth and Ax-to-Grind Urdu) are publicly available but state no licence; for these we distribute a script that downloads the original files and rebuilds our splits, unless the authors grant permission to redistribute. Generated domains are released with attribution to xAI Grok, as its terms of service request. The three untraced domains have no identifiable licence; we release them for research use and will remove any of them on request from a rights holder. The splits, build scripts, audits, model predictions and prompts are available at https://github.com/AbdullahPatti/Urdu-Multi-Domain-Script-Research and on Zenodo.

4 METHODOLOGY

4.1 Evaluation design

For every matrix with domains $D_1, \ldots, D_n$ we train (or prompt) one model per source domain D_i and evaluate it on the test split of every domain D_j of the matrix, filling an $n \times n$ matrix of macro-F1 scores $F(i \to j)$. The diagonal gives in-domain performance and the off-diagonal cells give the 80 cross-domain shifts across the seven matrices. Transfer is evaluated only inside a matrix, where the label space is identical, and scores are never pooled across scripts.

4.2 Metrics

All scores are macro-averaged F1 over the C classes of the matrix, $\mathrm{F1}_{\mathrm{macro}} = \frac{1}{C}\sum_c \frac{2P_cR_c}{P_c+R_c}$, which weights minority classes equally and is needed for the imbalanced abusive-language and fake-news domains. Following Calderon et al. [13], for a shift (i, j):

$$\mathrm{SS}_i = F(i \to i), \qquad \mathrm{TT}_j = F(j \to j), \tag{1}$$

$$\mathrm{ST}_{ij} = F(i \to j), \tag{2}$$

$$\Delta_S(i, j) = \mathrm{SS}_i - \mathrm{ST}_{ij}, \tag{3}$$

$$\Delta_T(i, j) = \mathrm{TT}_j - \mathrm{ST}_{ij}. \tag{4}$$

The Source Drop Δ_S measures how much a model loses when it is moved away from its own domain; the Target Drop Δ_T measures how far it falls short of a model trained on the target itself. A shift with $\Delta_S > 0$ but $\Delta_T \leq 0$ is a move to a *harder* target: the source model does as well there as the target's own model, so the loss reflects target difficulty rather than a robustness failure.

Averaged over all $n(n-1)$ shifts of a complete matrix, $\overline{\Delta_S} = \overline{\Delta_T}$ exactly, because each domain is a source in $n-1$ shifts and a target in $n-1$ shifts. A matrix-level mean Target Drop therefore adds nothing to the mean Source Drop. The Target Drop is informative *per shift* and *per target domain*, and we report it in three ways: the worst-case drops over shifts, $\mathrm{WSD} = \max_{i \neq j} \Delta_S(i, j)$ and $\mathrm{WTD} = \max_{i \neq j} \Delta_T(i, j)$; the percentage of shifts with a positive Target Drop and the percentage of harder-target shifts; and, per target domain j, TT_j with the mean Δ_T over all sources (Tables 11 and 12).

Relative robustness. $\%\mathrm{Rob} = 100 \cdot \overline{\mathrm{ST}}/\overline{\mathrm{SS}}$ is the share of in-domain performance retained across domains. Drops are reported in percentage points (pp) of macro-F1, and every claim about %Rob is accompanied by the absolute ST it is computed from.

4.3 Baselines and degeneracy guards

Every results table includes a majority-class baseline (the macro-F1 of always predicting the target's most frequent class) and a uniform-random baseline (its expected macro-F1, $\frac{1}{C}\sum_c \frac{2p_c/C}{p_c+1/C}$ for class priors p_c), each averaged over the target domains. It also includes a *length-only* shortcut baseline: a logistic regression on the logarithms of an item's character and word counts, trained on each source domain and evaluated on every target exactly like the models. In a matrix in which this baseline performs well, labels are partly predictable from length, and a model's advantage over it is the part of its performance that length cannot explain.

A model near the floor of the metric has little to lose, so its %Rob approaches 100% without reflecting robustness. Two guards address this. First, %Rob is suppressed (n/a†) when $\overline{\mathrm{SS}}$ is less than 0.05 above the uniform-random baseline, and whenever $\%\mathrm{Rob} \geq 100\%$. Second, %Rob is marked ‡ when $\overline{\mathrm{SS}}$ does not exceed the in-domain score of the length-only baseline, because such a model has not shown that it learned more than text length. When both guards apply, both marks are shown. Like every aggregate, %Rob is computed per seed (as $\overline{\mathrm{ST}}/\overline{\mathrm{SS}}$ of that seed) and then averaged, and

the guards are applied to these averages. The mean of the per-seed ratios can therefore reach 100% although the mean ST shown in a table is slightly below the mean SS.

4.4 Fine-tuned encoders

We fine-tune XLM-R base (xlm-roberta-base, 278M parameters) [14] and mBERT (bert-base-multilingual-cased, 178M) [17] with a linear classification head using Hugging Face Transformers [51]. Training data per source is capped at 8,000 examples by stratified sampling (fixed across seeds), and 10% is held out for validation. We use AdamW with learning rate 2×10^{-5}, linear warm-up over 10% of steps, weight decay 0.01, batch size 16, at most 5 epochs, and early stopping with patience 2 on validation macro-F1, keeping the best epoch. Inputs are truncated to 128 tokens, or 256 for fake news and QA; QA pairs are encoded as sentence pairs. Each configuration is run with three seeds (13, 42, 87), which vary initialisation, data order and the validation split [18, 43]. Training depends only on the source domain, so each (model, seed, source) is trained once and evaluated on every target of its matrix. Hyperparameters were fixed in advance and not tuned per domain.

4.5 Few-shot LLMs

We evaluate Llama-3.1-8B-Instruct [23], Qwen2.5-7B-Instruct [45] and Mistral-7B-Instruct-v0.3 [29], loaded with 4-bit NF4 quantisation [16]. The prompt (Appendix D), formatted with each model's chat template, states the task and script, lists the allowed labels, and gives five demonstrations per class sampled from the source training split and shuffled; the test item follows. Each text in the prompt is truncated to the same token budget as the encoders, so the test item is never cut from the prompt. Decoding is greedy with at most six new tokens. The prediction is the earliest whole-word label name in the output; outputs that name no label are counted as errors and reported as the parse-failure rate. Every configuration is repeated with three demonstration draws (seeds 13, 42, 87). LLMs are evaluated on a fixed stratified sample of up to 300 test items per target domain, identical across sources, models and seeds; whenever the two families are compared directly, fine-tuned predictions are restricted to the same items.

In-domain means something different for the two families. For a fine-tuned model, SS versus ST contrasts learned weights applied inside and outside their training distribution. For a few-shot LLM the weights never change: SS versus ST contrasts demonstrations drawn from the target domain with demonstrations drawn from another domain. The two relative-robustness figures therefore answer different questions (sensitivity to the training data versus sensitivity to the source of the demonstrations), and the two families also see very different amounts of source data (up to 8,000 training items against ten or fifteen demonstrations). Following Calderon et al. [13], we compare them as two practical ways of building a classifier from source-domain data, not as a like-for-like measurement of one mechanism.

4.6 Variance and significance

Every aggregate is computed per seed and reported as mean ± standard deviation over the three seeds (or demonstration draws). Differences between a fine-tuned model and an LLM are tested with a paired bootstrap [19, 34]: test items are resampled with replacement within each target domain, the same resample is applied to both systems, and mean ST (averaged over seeds and shifts) is recomputed 5,000 times. We report 95% confidence intervals and two-sided p-values, and because the 42 comparisons form one family of tests, we control the family-wise error rate with the Holm–Bonferroni procedure. A difference is called significant only when its Holm-adjusted p

is below 0.05. Differences between models of the same family are not tested and are described only as numbers.

4.7 Control for synthetic data

To test whether generated domains drive the conclusions (RQ4), we recompute every aggregate on the submatrix of *organic* domains (native Urdu text from public corpora, excluding generated, machine-translated and uncertain domains) for each matrix that retains at least two of them.

5 EXPERIMENTAL SETUP

All experiments ran on one workstation: Intel Core i7-13700 (16 cores), 32 GB RAM, NVIDIA RTX A4000 (16 GB), PyTorch 2.14 with CUDA 12.6, Transformers 4.57 and bitsandbytes 0.50. The evaluation comprises 156 fine-tuning runs (two encoders, three seeds, 26 source domains), which produce 636 scored matrix cells, and 954 LLM matrix cells (three models, three demonstration draws). Fine-tuning took 3.2 GPU-hours of training and LLM inference 4.0 GPU-hours. To make LLM inference feasible, the prompt prefix shared by all test items of a (source, demonstration draw) pair is encoded once and its attention cache reused. On a check of 32 fake-news items per model, predictions with and without the cache agreed on all 96 items.

6 RESULTS

Figure 2 summarises in-domain and cross-domain scores for every model and matrix; Tables 4–10 give the full aggregates, Figures 3 and 4 the transfer matrices, and Figure 6 the paired tests between the two families.

6.1 Nastaliq track

Sentiment (Table 4). The encoders reach SS of 0.813 ± 0.011 (XLM-R) and 0.805 ± 0.003 (mBERT) and keep 70.5–78.1% of it across domains (ST 0.567–0.636). The LLMs score slightly lower in-domain (SS 0.691–0.792) but lose almost nothing (ST 0.679–0.790). XLM-R's cross-domain score varies across seeds (SD 0.074), mostly on shifts into the translated movie reviews.

Hate speech (Table 5). The encoders score SS 0.897–0.903 but ST only 0.519–0.565, a mean Source Drop of 33.8–37.8 pp and a worst single-shift Target Drop of 70.8 pp for XLM-R. The LLMs are weaker in-domain (SS 0.651–0.750) and reach ST 0.542–0.581. Across domains XLM-R does not differ significantly from any LLM, whereas mBERT is significantly below all three, by −2.8 to −6.7 pp. Apart from QA, hate speech is the Nastaliq matrix with the smallest difference between the families.

Fake news (Table 6). Fine-tuned encoders fit each fake-news domain (SS 0.782 ± 0.119 for XLM-R, 0.860 ± 0.008 for mBERT) but transfer below chance: ST is 0.356–0.384, against a uniform-random baseline of 0.499 and a length-only cross-domain score of 0.354. XLM-R's high seed variance comes from FN-A, whose training split has 634 items. Among the LLMs only Llama-3.1-8B clearly exceeds both the random and the length-only baseline in-domain (SS 0.605 against 0.570). Qwen2.5-7B matches the length-only baseline (SS 0.570, above it by less than 0.001) and keeps all of it across domains: in one demonstration draw its ST exceeds its SS, so the mean of its per-draw ratios is just above 100% and its %Rob is suppressed. Mistral-7B is within 0.05 of random and below the length-only baseline, so it carries both marks. Llama-3.1-8B's cross-domain score (0.581) is significantly higher than that of both encoders (Figure 6), but it is still close to chance. Neither family provides a fake-news classifier that transfers between these two corpora.

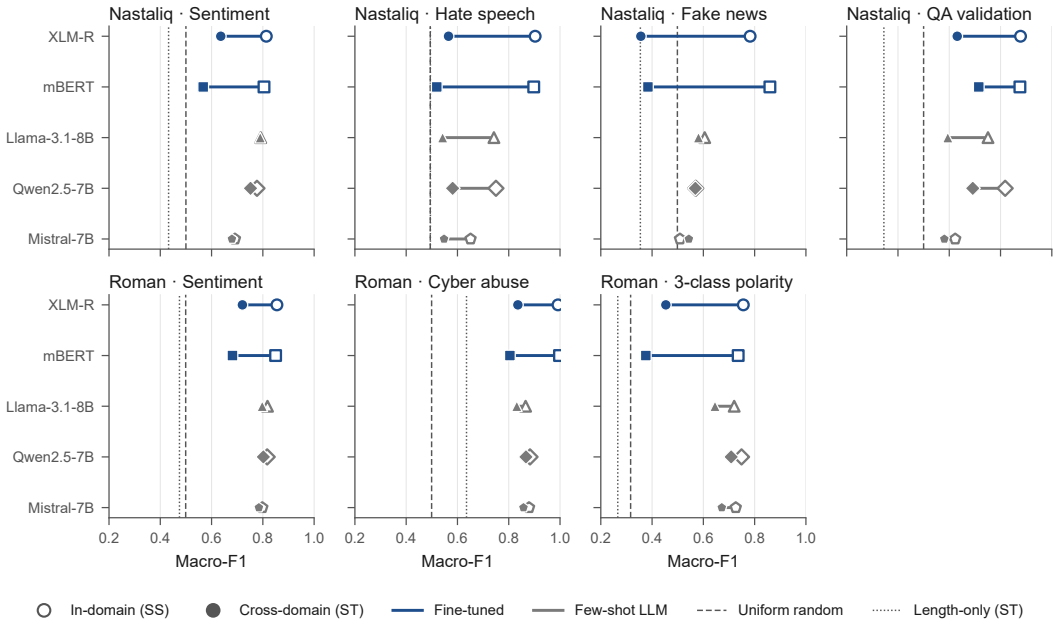


Fig. 2. In-domain (open markers) and cross-domain (filled markers) macro-F1 for every model and matrix, as means over three seeds or demonstration draws; the line between the two markers is the mean Source Drop. Blue: fine-tuned encoders; orange: few-shot LLMs; marker shape identifies the model. Seed standard deviations are in Tables 4–10. Dashed line: uniform-random baseline; dotted line: cross-domain score of the length-only baseline.

QA pair validation (Table 7). QA is the one matrix in which fine-tuning also wins across domains. The encoders reach SS 0.876–0.878 and ST 0.631–0.715, and both exceed Llama-3.1-8B (ST 0.594) and Mistral-7B (ST 0.581) significantly, by +3.5 to +13.0 pp. Qwen2.5-7B (SS 0.818, ST 0.692) is significantly above XLM-R (−6.2 pp, $p_{\text{Holm}} = 0.017$) and does not differ significantly from mBERT. The length-only baseline is at chance (SS 0.422), as expected for balanced pairs whose distractors come from the same domain.

6.2 Roman Urdu track

Sentiment, binary (Table 8). The encoders are close in-domain (SS 0.855 and 0.850) and lose 13.5–16.8 pp across domains (ST 0.682–0.720). The LLMs are slightly weaker in-domain (SS 0.798–0.818) and lose 1.3–2.0 pp, so their ST (0.785–0.802) is significantly higher than that of both encoders in all six comparisons (−6.4 to −12.9 pp).

Cyber abuse (Table 9). In-domain scores are close to perfect for the encoders (SS 0.992 and 0.998), and ST is 0.804–0.836. No test item has a near-duplicate in training, so the high in-domain scores are not caused by leakage; they indicate that these domains are easy to fit, which is consistent with the uniform style of the three generated domains and the augmentation-like noise of CA-A. The length-only baseline reaches SS 0.770 and ST 0.635, the highest of any matrix. XLM-R does not differ significantly from any LLM across domains; mBERT is significantly below Qwen2.5-7B and Mistral-7B.

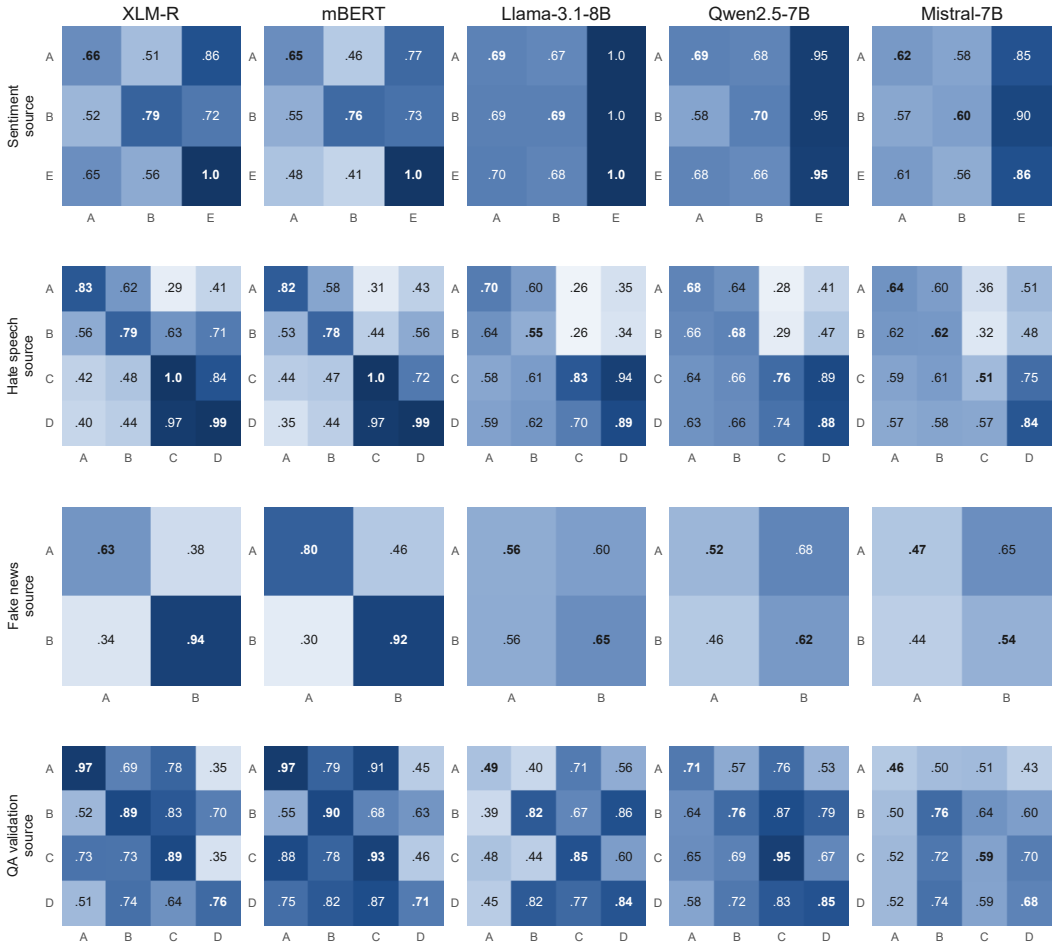


Fig. 3. Nastaliq transfer matrices: macro-F1 of a model trained or prompted on the row domain and tested on the column domain, mean over three seeds or demonstration draws. The diagonal (bold) is in-domain performance; darker cells are higher.

3-class polarity (Table 10). This matrix has the largest fine-tuned losses in the Roman track: SS 0.736–0.756, ST 0.376–0.454, a mean Source Drop of 30.2–36.0 pp and a worst Target Drop of 80.3 pp (XLM-R). The LLMs have similar in-domain scores (SS 0.720–0.749) but keep them (ST 0.645–0.708), and their advantage over the encoders is the largest in the study, from −19.4 to −33.5 pp, significant in all six comparisons. PR-B, which fails the per-class minimum (Section 3.3), is a source or target in 10 of the 30 shifts. Without it (lower block of Table 10), every score rises because PR-B is the hardest domain to fit: the encoders reach SS 0.802–0.848 and ST 0.411–0.523, and the LLMs SS 0.762–0.789 and ST 0.682–0.764. The fine-tuned Source Drop stays at 32.5–39.1 pp against 2.5–8.0 pp for the LLMs, and the share of harder-target shifts of the encoders falls to 0–0%. The comparison between the families does not depend on PR-B.

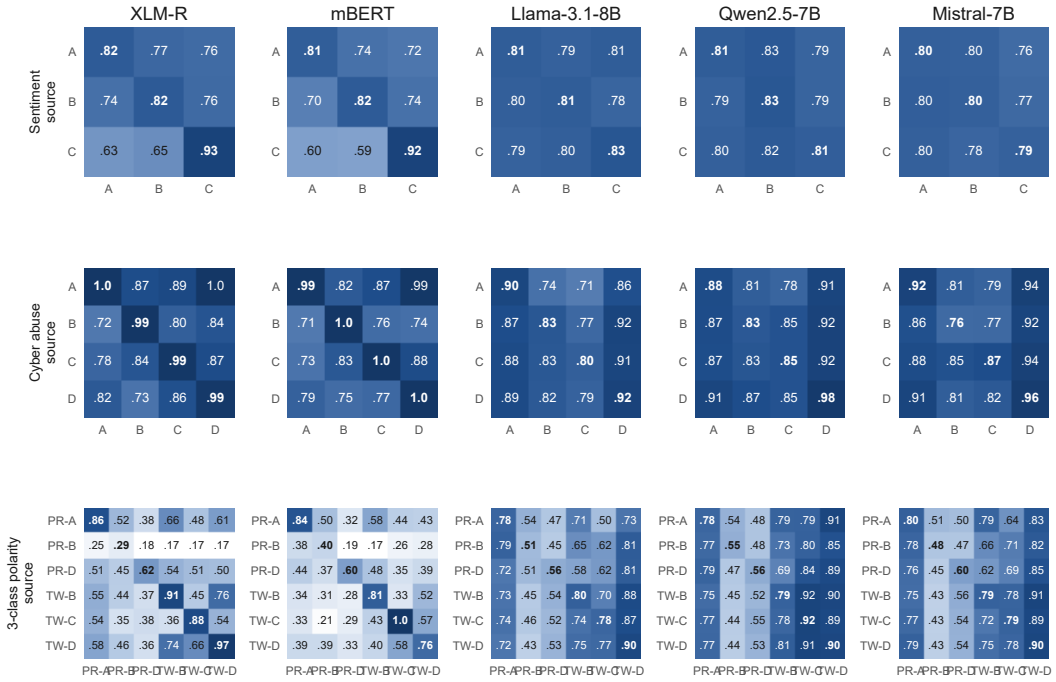


Fig. 4. Roman Urdu transfer matrices; layout as in Figure 3.

Table 4. Nastaliq Sentiment (3 domains, 6 shifts). Macro-F1, mean±SD over 3 seeds (fine-tuned) or 3 demonstration draws (LLMs). $\overline{\Delta_S}$, WSD and WTD in pp; $\Delta_T{>}0$ %: share of shifts with positive Target Drop; Harder-tgt %: share with $\Delta_S > 0$ and $\Delta_T \leq 0$. †: %Rob suppressed (SS within 0.05 of random, or %Rob ≥ 100); ‡: SS not above the length-only baseline; both marks are shown when both apply (Section 4.3). %Rob and the guards use the mean over seeds of the per-seed ratio ST/SS. Drops and %Rob are computed from unrounded scores, so they can differ by 0.1 from differences of the displayed values.

Model	SS	ST	$\overline{\Delta_S}$	WSD	WTD	$\Delta_T{>}0$ %	Harder-tgt %	%Rob
Majority class	0.336	–	–	–	–	–	–	–
Uniform random	0.500	–	–	–	–	–	–	–
Length-only (shortcut)	0.552	0.432	11.9	27.0	21.3	100.0	0.0	78.4
XLM-R (fine-tuned)	0.813 ± 0.011	0.636 ± 0.074	17.8	43.3	33.7	94.4	5.6	78.1
mBERT (fine-tuned)	0.805 ± 0.003	0.567 ± 0.008	23.7	58.8	35.3	100.0	0.0	70.5
Llama-3.1-8B (few-shot)	0.792 ± 0.024	0.790 ± 0.010	0.2	32.2	3.4	44.4	22.2	99.8
Qwen2.5-7B (few-shot)	0.777 ± 0.011	0.752 ± 0.009	2.5	28.7	10.6	72.2	0.0	96.8
Mistral-7B (few-shot)	0.691 ± 0.019	0.679 ± 0.018	1.2	30.2	6.5	66.7	11.1	98.3

7 ANALYSIS AND DISCUSSION

7.1 Robustness failures or harder targets? (RQ1)

Figure 5 plots the Source Drop of every shift against its Target Drop. For the fine-tuned encoders, 83–100% of shifts per matrix have a positive Target Drop, and at most 17% are harder-target shifts. Their Source Drops of 13.5–47.6 pp are therefore not explained by moving to difficult domains: in almost every shift, a model trained on the target does better there.

Table 5. Nastaliq Hate Speech (4 domains, 12 shifts). Columns as in Table 4.

Model	SS	ST	$\overline{\Delta_S}$	WSD	WTD	$\Delta_T>0$ %	Harder-tgt %	%Rob
Majority class	0.372	–	–	–	–	–	–	–
Uniform random	0.494	–	–	–	–	–	–	–
Length-only (shortcut)	0.595	0.494	10.0	24.6	29.2	83.3	8.3	83.1
XLM-R (fine-tuned)	0.903 ± 0.004	0.565 ± 0.002	33.8	58.9	70.8	100.0	0.0	62.6
mBERT (fine-tuned)	0.897 ± 0.003	0.519 ± 0.007	37.8	64.9	69.2	100.0	0.0	57.9
Llama-3.1-8B (few-shot)	0.742 ± 0.027	0.542 ± 0.002	20.0	44.3	59.5	80.6	13.9	73.1
Qwen2.5-7B (few-shot)	0.750 ± 0.007	0.581 ± 0.005	16.9	40.4	50.2	88.9	5.6	77.5
Mistral-7B (few-shot)	0.651 ± 0.006	0.548 ± 0.017	10.3	32.3	36.4	80.6	13.9	84.1

Table 6. Nastaliq Fake News (2 domains, 2 shifts). Columns as in Table 4.

Model	SS	ST	$\overline{\Delta_S}$	WSD	WTD	$\Delta_T>0$ %	Harder-tgt %	%Rob
Majority class	0.353	–	–	–	–	–	–	–
Uniform random	0.499	–	–	–	–	–	–	–
Length-only (shortcut)	0.570	0.354	21.5	26.9	29.3	100.0	0.0	62.2
XLM-R (fine-tuned)	0.782 ± 0.119	0.356 ± 0.030	42.6	60.0	55.9	100.0	0.0	45.9
mBERT (fine-tuned)	0.860 ± 0.008	0.384 ± 0.005	47.6	61.8	49.5	100.0	0.0	44.7
Llama-3.1-8B (few-shot)	0.605 ± 0.061	0.581 ± 0.022	2.4	9.1	7.6	50.0	33.3	96.4
Qwen2.5-7B (few-shot)	0.570 ± 0.054	0.569 ± 0.021	0.1	15.5	6.3	66.7	0.0	n/a†
Mistral-7B (few-shot)	0.509 ± 0.042	0.543 ± 0.023	-3.5	10.6	3.7	50.0	0.0	n/a†‡

Table 7. Nastaliq QA Pair Validation (4 domains, 12 shifts). Columns as in Table 4.

Model	SS	ST	$\overline{\Delta_S}$	WSD	WTD	$\Delta_T>0$ %	Harder-tgt %	%Rob
Majority class	0.334	–	–	–	–	–	–	–
Uniform random	0.500	–	–	–	–	–	–	–
Length-only (shortcut)	0.422	0.345	7.7	18.0	17.9	50.0	41.7	n/a†
XLM-R (fine-tuned)	0.878 ± 0.017	0.631 ± 0.010	24.7	62.0	53.4	94.4	2.8	71.9
mBERT (fine-tuned)	0.876 ± 0.008	0.715 ± 0.009	16.0	53.1	41.4	100.0	0.0	81.7
Llama-3.1-8B (few-shot)	0.751 ± 0.009	0.594 ± 0.059	15.7	46.4	44.3	80.6	13.9	79.2
Qwen2.5-7B (few-shot)	0.818 ± 0.013	0.692 ± 0.032	12.6	33.0	31.5	97.2	0.0	84.6
Mistral-7B (few-shot)	0.623 ± 0.020	0.581 ± 0.012	4.2	25.9	27.7	52.8	36.1	93.3

The per-target view (Tables 11 and 12) makes this concrete for hate speech and fake news. XLM-R trained on the generated HS-C or HS-D scores TT of 0.999 and 0.992 on its own domain, while models from the other sources fall short by 36.8 and 33.8 pp on average. These near-perfect scores do not mean that HS-C and HS-D are clean or natural targets. Both are partly templated (28% and 41% of items have a near-duplicate variant, Appendix Table 18), and in HS-C a single word separates the classes (Section 8), so a model can fit them by recognising a pattern. What the scores do show is that these targets are easy to fit, including by their own models, so the cross-domain failure on them is not caused by target difficulty. The organic fake-news target FN-B leads to the same conclusion without that caveat: XLM-R trained on FN-B scores 0.936 there, and the FN-A

Table 8. Roman Sentiment, binary (3 domains, 6 shifts). Columns as in Table 4.

Model	SS	ST	$\overline{\Delta_S}$	WSD	WTD	$\Delta_T>0$ %	Harder-tgt %	%Rob
Majority class	0.350	–	–	–	–	–	–	–
Uniform random	0.499	–	–	–	–	–	–	–
Length-only (shortcut)	0.600	0.475	12.5	24.3	21.1	100.0	0.0	79.1
XLM-R (fine-tuned)	0.855 ± 0.000	0.720 ± 0.005	13.5	29.7	18.7	100.0	0.0	84.2
mBERT (fine-tuned)	0.850 ± 0.003	0.682 ± 0.009	16.8	34.3	23.2	100.0	0.0	80.2
Llama-3.1-8B (few-shot)	0.818 ± 0.011	0.798 ± 0.006	2.0	4.4	5.0	88.9	5.6	97.5
Qwen2.5-7B (few-shot)	0.817 ± 0.015	0.802 ± 0.007	1.5	5.2	2.9	88.9	0.0	98.2
Mistral-7B (few-shot)	0.798 ± 0.021	0.785 ± 0.023	1.3	5.4	3.8	77.8	16.7	98.3

Table 9. Roman Cyber Abuse (4 domains, 12 shifts). Columns as in Table 4.

Model	SS	ST	$\overline{\Delta_S}$	WSD	WTD	$\Delta_T>0$ %	Harder-tgt %	%Rob
Majority class	0.350	–	–	–	–	–	–	–
Uniform random	0.499	–	–	–	–	–	–	–
Length-only (shortcut)	0.770	0.635	13.5	25.3	34.2	83.3	8.3	82.5
XLM-R (fine-tuned)	0.992 ± 0.008	0.836 ± 0.003	15.7	27.7	27.9	91.7	0.0	84.2
mBERT (fine-tuned)	0.998 ± 0.001	0.804 ± 0.014	19.4	28.9	28.5	100.0	0.0	80.6
Llama-3.1-8B (few-shot)	0.866 ± 0.023	0.831 ± 0.031	3.4	19.8	17.1	66.7	19.4	96.0
Qwen2.5-7B (few-shot)	0.883 ± 0.022	0.867 ± 0.017	1.6	13.4	9.6	58.3	27.8	98.2
Mistral-7B (few-shot)	0.879 ± 0.048	0.857 ± 0.053	2.2	16.5	12.4	80.6	16.7	97.5

model falls short by 55.9 pp. In both matrices the source models fail on targets that a model trained on the target fits well, which indicates that each model learns cues specific to its own source.

One target behaves differently. On Roman PR-B, XLM-R's own TT is 0.294, below the random baseline of the matrix, and models from other sources do better there (mean Δ_T -14.9 pp). Its test split has 8 negative items out of 148, and its training split is similarly skewed, so PR-B is a hard target for reasons of class balance rather than content; it fails the per-class minimum of Section 3.3 and accounts for most of the harder-target shifts of the fine-tuned models. Without PR-B, at most 6% of the fine-tuned shifts in any matrix are harder-target shifts.

For the LLMs the picture is flatter, with 44–97% of shifts with a positive Target Drop and up to 36% harder-target shifts. Their largest per-target drops fall on the generated hate-speech domains (HS-C: 41.9 pp for Llama-3.1-8B, 32.4 pp for Qwen2.5-7B), where demonstrations from the target itself are needed to convey what the domain counts as hateful (Section 7.6).

7.2 Fine-tuned encoders versus few-shot LLMs (RQ2)

Figure 6 and Table 13 give all 42 paired comparisons of cross-domain F1. After Holm correction the few-shot LLM is significantly better in 30, the fine-tuned encoder in 4, and 8 differences are not significant. Fine-tuned encoders are stronger in-domain than every LLM in 6 of the 7 matrices. This reproduces the pattern reported for English by Calderon et al. [13], with two qualifications. First, the four comparisons won by fine-tuning are all in QA validation, where the encoders read the question and answer as a sentence pair and generalise across QA domains better than Llama-3.1-8B and Mistral-7B. Second, the LLM advantage in fake news is an advantage over a below-chance transfer and does not make the LLMs usable there.

Table 10. Roman 3-class Polarity: reviews and social-media opinions (6 domains, 30 shifts). Columns as in Table 4. Lower block: the same aggregates without PR-B, which fails the per-class minimum of Section 3.3 (5 domains, 20 shifts); the majority and random rows refer to the full matrix.

Model	SS	ST	$\overline{\Delta_S}$	WSD	WTD	$\Delta_T>0$ %	Harder-tgt %	%Rob
Majority class	0.205	–	–	–	–	–	–	–
Uniform random	0.316	–	–	–	–	–	–	–
Length-only (shortcut)	0.385	0.266	11.8	28.4	37.7	90.0	0.0	69.2
XLM-R (fine-tuned)	0.756 ± 0.005	0.454 ± 0.020	30.2	62.0	80.3	83.3	16.7	60.0
mBERT (fine-tuned)	0.736 ± 0.011	0.376 ± 0.016	36.0	79.5	75.4	93.3	6.7	51.0
Llama-3.1-8B (few-shot)	0.720 ± 0.011	0.645 ± 0.023	7.5	46.7	28.7	87.8	7.8	89.6
Qwen2.5-7B (few-shot)	0.749 ± 0.009	0.708 ± 0.003	4.0	47.2	13.1	80.0	11.1	94.6
Mistral-7B (few-shot)	0.726 ± 0.003	0.672 ± 0.007	5.4	47.2	19.3	90.0	5.6	92.5
Without PR-B (5 domains, 20 shifts)								
Length-only (shortcut)	0.397	0.273	12.4	28.4	37.7	95.0	0.0	68.9
XLM-R (fine-tuned)	0.848 ± 0.006	0.523 ± 0.028	32.5	61.1	59.9	100.0	0.0	61.7
mBERT (fine-tuned)	0.802 ± 0.009	0.411 ± 0.009	39.1	72.1	69.7	100.0	0.0	51.2
Llama-3.1-8B (few-shot)	0.762 ± 0.008	0.682 ± 0.026	8.0	37.7	28.7	91.7	5.0	89.4
Qwen2.5-7B (few-shot)	0.789 ± 0.006	0.764 ± 0.002	2.5	37.8	12.5	71.7	15.0	96.8
Mistral-7B (few-shot)	0.777 ± 0.003	0.724 ± 0.005	5.3	36.5	17.6	90.0	3.3	93.2

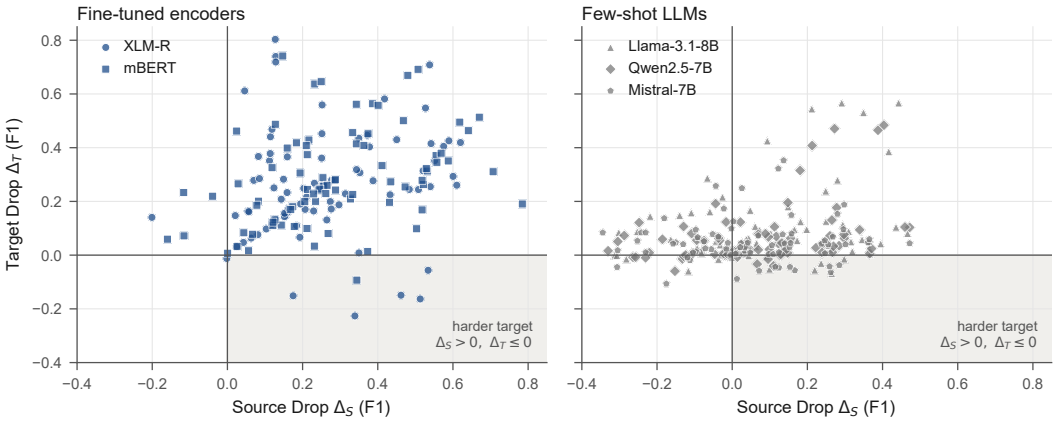


Fig. 5. Source Drop against Target Drop for every shift (one point per shift and model, means over seeds). Points in the shaded lower-right region are harder-target shifts: the source model loses performance but matches or beats the target's own model.

Are low LLM scores a prompting artefact? Parse failures are rare (Table 14): at most 0.32% of items for Llama-3.1-8B and Qwen2.5-7B in any matrix. Mistral-7B is the exception on the two binary sentiment matrices, with 8.02% (Nastaliq) and 4.54% (Roman) unparseable outputs. These outputs are almost all "Neutral", often followed by a justification: the model declines to choose between the two allowed labels rather than producing malformed text. They count as errors, so Mistral-7B's sentiment scores are a lower bound for this prompt. Elsewhere, low LLM scores are not caused by unparseable outputs.

Table 11. Per-target in-domain score TT_j and mean Target Drop $\overline{\Delta_T}$ (pp, mean over all source domains) for the Nastaliq matrices. A negative $\overline{\Delta_T}$ means that models from other domains do better on the target than its own model.

Target	XLM-R		mBERT		Llama-3.1-8B		Qwen2.5-7B		Mistral-7B	
	TT	$\overline{\Delta_T}$	TT	$\overline{\Delta_T}$	TT	$\overline{\Delta_T}$	TT	$\overline{\Delta_T}$	TT	$\overline{\Delta_T}$
Sentiment										
SA-A	0.655	7.0	0.650	13.4	0.685	-1.1	0.689	5.8	0.618	2.6
SA-B	0.789	25.4	0.763	32.8	0.692	1.7	0.696	2.5	0.597	2.7
SA-E	0.997	20.9	1.000	24.9	0.998	-0.1	0.946	-0.8	0.859	-1.7
Hate Speech										
HS-A	0.829	36.6	0.816	37.4	0.704	10.1	0.680	3.4	0.637	4.5
HS-B	0.792	27.9	0.781	28.6	0.552	-5.9	0.683	2.9	0.620	2.3
HS-C	0.999	36.8	1.000	42.9	0.828	41.9	0.760	32.4	0.510	9.3
HS-D	0.992	33.8	0.993	42.4	0.885	33.9	0.878	28.8	0.836	25.3
Fake News										
FN-A	0.629	29.3	0.798	49.5	0.558	-0.3	0.524	6.3	0.474	3.7
FN-B	0.936	55.9	0.921	45.6	0.653	5.2	0.617	-6.0	0.543	-10.7
QA Validation										
QA-A	0.965	37.7	0.968	23.9	0.489	5.0	0.714	8.8	0.460	-5.2
QA-B	0.891	16.9	0.897	9.7	0.821	27.1	0.761	10.4	0.761	10.7
QA-C	0.891	14.1	0.928	10.7	0.852	13.6	0.949	13.2	0.590	1.2
QA-D	0.765	30.0	0.709	19.9	0.842	17.0	0.848	18.2	0.683	10.3

7.3 Label granularity: binary versus three-class polarity

Roman Sentiment and Roman 3-class polarity are the same task type with different label sets, and the label set accounts for much of the difference between their scores. The random baseline is 0.499 for the binary matrix and 0.316 for the three-class matrix. XLM-R scores SS 0.855 on binary sentiment against 0.756 on three-class polarity, and its mean Source Drop rises from 13.5 to 30.2 pp. The LLMs, whose in-domain scores fall by a similar amount, keep their Source Drop small in both matrices (1.3–2.0 and 4.0–7.5 pp).

Inside the three-class matrix, the collection of origin matters as much as the label set. XLM-R reaches a mean ST of 0.585 on shifts between the three generated opinion domains, 0.435 between reviews and opinions, and 0.380 between the three organic review domains. The generated opinion domains share a generator and transfer well among themselves; the organic review domains (e-commerce, restaurants and general comments) differ in platform, class balance and the amount of English, and transfer worst. The LLMs show the same order (Qwen2.5-7B: 0.867, 0.696 and 0.587) at a higher level.

Table 12. As Table 11, for the Roman Urdu matrices.

Target	XLM-R		mBERT		Llama-3.1-8B		Qwen2.5-7B		Mistral-7B	
	TT	$\overline{\Delta_T}$	TT	$\overline{\Delta_T}$	TT	$\overline{\Delta_T}$	TT	$\overline{\Delta_T}$	TT	$\overline{\Delta_T}$
Sentiment (binary)										
SA-A	0.817	13.2	0.807	16.0	0.811	1.4	0.812	1.6	0.799	-0.3
SA-B	0.822	10.9	0.817	15.2	0.809	1.1	0.833	0.9	0.800	1.2
SA-C	0.926	16.3	0.925	19.3	0.834	3.7	0.806	2.0	0.795	3.1
Cyber Abuse										
CA-A	0.997	22.3	0.993	24.7	0.904	2.7	0.878	-0.8	0.923	4.0
CA-B	0.995	18.1	1.000	19.9	0.834	3.7	0.828	-0.9	0.765	-5.9
CA-C	0.991	14.0	0.999	20.0	0.802	4.5	0.849	2.2	0.866	7.3
CA-D	0.986	8.1	1.000	12.9	0.923	2.9	0.978	5.8	0.962	3.2
3-class Polarity										
PR-A	0.860	37.3	0.841	46.5	0.779	3.6	0.777	0.7	0.798	2.8
PR-B	0.294	-14.9	0.404	4.9	0.511	3.1	0.548	7.8	0.475	2.6
PR-D	0.621	28.9	0.601	31.8	0.557	5.5	0.556	4.4	0.598	7.6
TW-B	0.905	41.3	0.808	39.6	0.799	11.2	0.793	3.4	0.793	8.7
TW-C	0.885	43.2	0.998	60.6	0.778	13.8	0.917	6.4	0.790	6.9
TW-D	0.970	45.5	0.763	32.7	0.897	7.6	0.903	1.6	0.904	4.1

7.4 Organic-only analysis (RQ4)

Table 15 repeats the analysis on the organic submatrices. Nastaliq sentiment (one organic domain) and Roman cyber abuse (none) drop out; fake news and Roman binary sentiment are organic throughout and are unchanged. The answer to RQ2 holds: in every remaining matrix the fine-tuned encoders retain less of their in-domain score than the LLMs, except in QA, where Llama-3.1-8B retains only 59.7% (ST 0.390). Fake news remains the matrix with the lowest fine-tuned retention (45.9% for XLM-R) and binary sentiment the highest (84.2%). One result changes. On its two organic domains, hate speech is no longer among the least robust matrices for fine-tuned models (73.0% against 62.6% on all four domains), because its largest drops involve the generated HS-C and HS-D. Part of the low hate-speech robustness of the full matrix is thus a property of the generated domains. The organic analysis reverses the in-domain order of the two encoders in one place only: on the organic 3-class submatrix mBERT scores higher than XLM-R (SS 0.615 against 0.592), the opposite of the full matrix (0.736 against 0.756). The reversal comes from PR-B, one of only three organic domains there, on which XLM-R's own score is 0.294 against 0.404 for mBERT; on the other two organic domains, PR-A and PR-D, XLM-R is the stronger encoder in-domain (Table 12). Differences within a family are not tested (Section 4.6), and this one reflects a single imbalanced domain rather than a difference between the encoders.

7.5 Nastaliq versus Roman Urdu

The two tracks share only two task types (sentiment and abusive language) and differ in sources and registers, so script effects cannot be separated from source effects, and we compare them

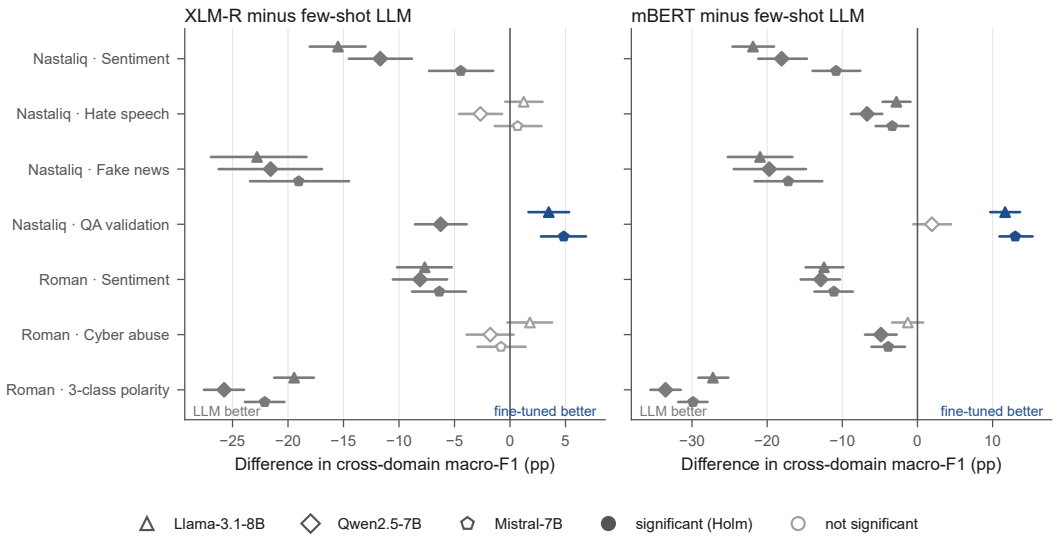


Fig. 6. Difference in mean cross-domain macro-F1 between each fine-tuned encoder and each LLM, with 95% paired bootstrap confidence intervals (5,000 resamples). Filled markers: significant after Holm correction over all 42 comparisons; orange: the LLM is better; blue: the fine-tuned encoder is better; open grey: not significant.

Table 13. Paired bootstrap comparison of mean cross-domain F1 (ST), fine-tuned encoder minus few-shot LLM, in pp with 95% confidence intervals (5,000 resamples of test items per target, shared by both systems; fine-tuned predictions restricted to the LLM evaluation sample). Negative values favour the LLM. *: Holm-adjusted $p < 0.05$ over all 42 comparisons.

Matrix	Fine-tuned	− Llama-3.1-8B	− Qwen2.5-7B	− Mistral-7B
Sentiment	XLM-R	-15.5* [-18.0, -13.0]	-11.7* [-14.5, -8.8]	-4.4* [-7.3, -1.5]
	mBERT	-21.9* [-24.6, -19.1]	-18.1* [-21.2, -14.7]	-10.8* [-14.0, -7.6]
Hate Speech	XLM-R	+1.2 [-0.4, +2.9]	-2.7 [-4.6, -0.7]	+0.7 [-1.4, +2.8]
	mBERT	-2.8* [-4.6, -0.9]	-6.7* [-8.8, -4.7]	-3.4* [-5.6, -1.2]
Fake News	XLM-R	-22.8* [-26.9, -18.3]	-21.6* [-26.2, -16.9]	-19.0* [-23.4, -14.5]
	mBERT	-20.9* [-25.2, -16.6]	-19.7* [-24.4, -14.8]	-17.2* [-21.7, -12.7]
QA Validation	XLM-R	+3.5* [+1.7, +5.3]	-6.2* [-8.6, -3.9]	+4.8* [+2.8, +6.8]
	mBERT	+11.7* [+9.7, +13.6]	+1.9 [-0.6, +4.5]	+13.0* [+10.9, +15.3]
Sentiment (binary)	XLM-R	-7.7* [-10.2, -5.2]	-8.1* [-10.6, -5.7]	-6.4* [-8.8, -4.0]
	mBERT	-12.4* [-14.9, -9.9]	-12.9* [-15.5, -10.3]	-11.1* [-13.7, -8.6]
Cyber Abuse	XLM-R	+1.8 [-0.2, +3.8]	-1.8 [-3.9, +0.3]	-0.8 [-2.9, +1.4]
	mBERT	-1.3 [-3.4, +0.8]	-4.9* [-7.0, -2.8]	-3.9* [-6.1, -1.7]
3-class Polarity	XLM-R	-19.4* [-21.2, -17.7]	-25.8* [-27.6, -24.0]	-22.1* [-23.9, -20.3]
	mBERT	-27.2* [-29.1, -25.2]	-33.5* [-35.5, -31.5]	-29.9* [-31.8, -27.9]

only descriptively. For sentiment, fine-tuned retention is 70.5–78.1% in Nastaliq and 80.2–84.2% in Roman Urdu. For abusive language it is lower in Nastaliq (57.9–62.6%) than in Roman Urdu (80.6–84.2%), but the Nastaliq matrix combines two organic tweet corpora with two generated domains that label mild criticism as hateful (Section 7.6), while three of the four Roman domains

Table 14. LLM parse-failure rate (% of all evaluated items, over all shifts and demonstration draws): outputs that name none of the allowed labels. They are scored as errors.

Matrix	Llama-3.1-8B	Qwen2.5-7B	Mistral-7B
Sentiment (N)	0.03	0.32	8.02
Hate Speech (N)	0.00	0.00	0.02
Fake News (N)	0.00	0.00	0.00
QA Validation (N)	0.00	0.00	0.00
Sentiment (binary) (R)	0.07	0.12	4.54
Cyber Abuse (R)	0.00	0.00	0.00
3-class Polarity (R)	0.15	0.00	0.00

are generated with a shared style. These data do not support a claim that either script is harder to transfer across.

7.6 What causes the largest failures? (RQ3)

For each matrix we examine the *worst-transferring shift*: the source→target pair with the largest Target Drop for XLM-R, so that the target's own model does well and the failure cannot be attributed to a hard target. Figure 7 shows confusion matrices for that shift, and Table 16 breaks down every XLM-R error on it. An item counts as an error when at least two of the three seeds misclassify it. Instead of labelling errors by hand, we measure three properties of each error and of each correctly classified item of the same target: whether its length lies outside the 5th–95th percentile of the source training split, whether more than half of its word types never occur in the source training split, and whether the target's own XLM-R also misclassifies it. A property that is much more frequent among errors than among correct items is a candidate explanation.

Collapse to one class. In the worst shifts of three matrices the source model assigns one class to nearly every target item (Figure 7). All 216 errors of the HS-A model on HS-C are hate items labelled *normal*; all 772 errors of the FN-A model on FN-B are fake items labelled *real*; and the PR-B model predicts *positive* for every TW-D item. In two further matrices one class dominates the predictions: the Nastaliq tweet model labels most translated movie reviews *negative*, and the QA-D model accepts most invalid QA-A pairs. For PR-B the training data explain the collapse, since 79% of its items are positive. The few-shot LLM avoids the collapse in every case except hate speech, where it also labels almost all HS-C hate items as normal.

What the measured properties show. Vocabulary and length rarely separate errors from correct items. In the sentiment, fake-news and QA shifts most target items lie outside the source's length range whether they are classified correctly or not, so length describes the whole target rather than the errors. In hate speech and cyber abuse, errors and correct items do not differ on either property. The one clear signal is in Roman binary sentiment, where 29% of errors are also misclassified by the target's own model against 11% of correct items; many of these errors are items that are ambiguous or mislabelled in SA-A itself. Taken together, the worst failures reflect differences in what a label means and how often each class occurs, more than unfamiliar words.

Table 15. Organic-only analysis: aggregates recomputed on the submatrix of organic domains (listed in brackets) for matrices with at least two. Symbols as in Table 4.

Matrix (organic domains)	Model	SS	ST	$\overline{\Delta_S}$	%Rob
Hate Speech (HS-A, HS-B)	XLM-R	0.811 ± 0.008	0.591 ± 0.004	21.9	73.0
	mBERT	0.798 ± 0.003	0.554 ± 0.004	24.4	69.4
	Llama-3.1-8B	0.628 ± 0.050	0.622 ± 0.013	0.6	99.4
	Qwen2.5-7B	0.681 ± 0.014	0.654 ± 0.033	2.7	95.9
	Mistral-7B	0.629 ± 0.023	0.613 ± 0.034	1.5	97.5
Fake News (FN-A, FN-B)	XLM-R	0.782 ± 0.119	0.356 ± 0.030	42.6	45.9
	mBERT	0.860 ± 0.008	0.384 ± 0.005	47.6	44.7
	Llama-3.1-8B	0.605 ± 0.061	0.581 ± 0.022	2.4	96.4
	Qwen2.5-7B	0.570 ± 0.054	0.569 ± 0.021	0.1	n/a†
	Mistral-7B	0.509 ± 0.042	0.543 ± 0.023	-3.5	n/a†‡
QA Validation (QA-A, QA-B)	XLM-R	0.928 ± 0.006	0.605 ± 0.037	32.3	65.2
	mBERT	0.932 ± 0.006	0.672 ± 0.015	26.0	72.1
	Llama-3.1-8B	0.655 ± 0.032	0.390 ± 0.032	26.5	59.7
	Qwen2.5-7B	0.737 ± 0.014	0.605 ± 0.057	13.3	82.0
	Mistral-7B	0.610 ± 0.017	0.502 ± 0.057	10.8	82.5
Sentiment (binary) (SA-A, SA-B, SA-C)	XLM-R	0.855 ± 0.000	0.720 ± 0.005	13.5	84.2
	mBERT	0.850 ± 0.003	0.682 ± 0.009	16.8	80.2
	Llama-3.1-8B	0.818 ± 0.011	0.798 ± 0.006	2.0	97.5
	Qwen2.5-7B	0.817 ± 0.015	0.802 ± 0.007	1.5	98.2
	Mistral-7B	0.798 ± 0.021	0.785 ± 0.023	1.3	98.3
3-class Polarity (PR-A, PR-B, PR-D)	XLM-R	0.592 ± 0.003	0.380 ± 0.005	21.2	64.2
	mBERT	0.615 ± 0.013	0.367 ± 0.006	24.8	59.7
	Llama-3.1-8B	0.616 ± 0.021	0.581 ± 0.006	3.4	94.5
	Qwen2.5-7B	0.627 ± 0.018	0.587 ± 0.008	4.1	93.6
	Mistral-7B	0.624 ± 0.013	0.578 ± 0.013	4.6	92.7

Table 16. Measured breakdown of XLM-R errors on the worst-transferring shift of each matrix. Errors: items misclassified by at least two of three seeds, out of all test items of the target. Main confusion: most frequent true→predicted pair and its share of errors. The last three columns give the percentage of errors (err) and of correctly classified items (ok) with each property: length outside the source's 5th–95th percentile; more than half of word types unseen in the source training data; also misclassified by the target's own XLM-R.

Matrix	Shift	TT / ST	Errors	Main confusion (%)	Length shift err / ok	Unseen vocab. err / ok	Hard in-domain err / ok
Sentiment (N)	SA-A→SA-B	0.79 / 0.51	663/1,638	Positive→Negative (93)	100 / 100	3 / 1	22 / 17
Hate Speech (N)	HS-A→HS-C	1.00 / 0.29	216/340	Hate→Normal (100)	0 / 0	0 / 0	0 / 0
Fake News (N)	FN-A→FN-B	0.94 / 0.38	772/1,645	Fake→Real (100)	78 / 99	0 / 0	7 / 4
QA Validation (N)	QA-D→QA-A	0.97 / 0.51	110/250	Invalid→Valid (100)	100 / 100	99 / 100	5 / 1
Sentiment (binary) (R)	SA-C→SA-A	0.82 / 0.63	773/2,140	Negative→Positive (50)	22 / 16	49 / 45	29 / 11
Cyber Abuse (R)	CA-B→CA-A	1.00 / 0.72	234/866	Abusive→Normal (66)	20 / 21	44 / 48	0 / 0
3-class Polarity (R)	PR-B→TW-D	0.97 / 0.17	213/320	Negative→Positive (52)	0 / 0	95 / 83	5 / 0

Examples. Table 17 shows misclassified items drawn at random from those that all three seeds get wrong and the target's own model gets right (six candidates per shift). None of the six sampled

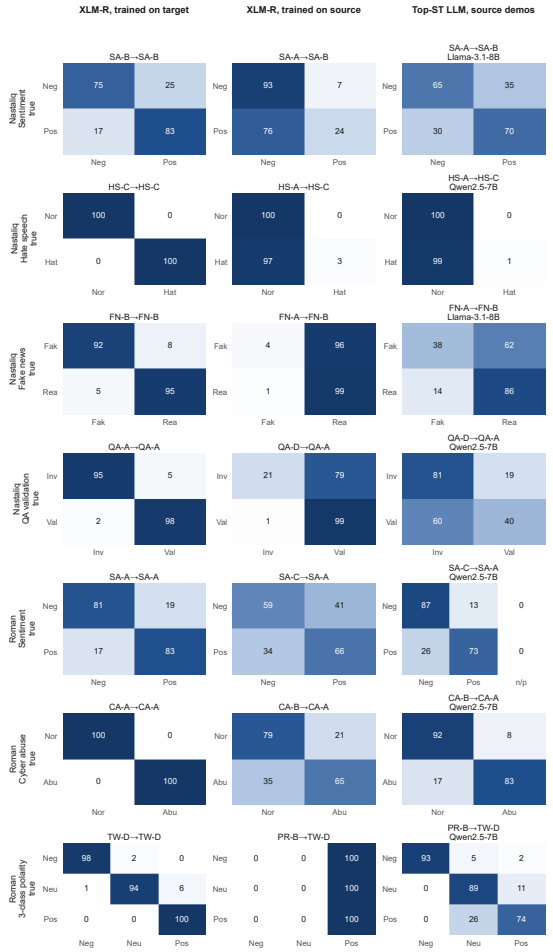


Fig. 7. Confusion matrices (row-normalised, %, summed over three seeds) for the worst-transferring shift of each matrix. Left: XLM-R trained on the target (in-domain reference). Middle: XLM-R trained on the source. Right: the few-shot LLM with the highest mean ST in the matrix, prompted with source demonstrations, on its evaluation sample; n/p = unparseable output. Darker cells hold more items.

HS-C errors contains a slur, insult or threat: they are mild criticism of leaders, players or online behaviour, labelled hateful by the generated HS-C domain. Hateful HS-A tweets, such as the example in Table 1, contain insults, so a model trained on HS-A, and an LLM given HS-A demonstrations, calls the HS-C items normal. The two domains share a label name but not a label definition. In fake news, the six sampled FN-B errors are short claims as they circulated on social media (a funeral presented as a protest, a map said to show a wrong border), whereas the fake items of FN-A are full fabricated articles of several hundred words written by journalists [3]. A model that learned what a fabricated article looks like in FN-A has no basis for judging a one-line claim. The Roman sentiment and polarity examples are unambiguous complaints predicted as positive, and in cyber abuse a friendly CA-A comment with a misspelling and an emoticon (*vidiyo, :)*) is predicted abusive, which is consistent with the augmentation-like noise that led us to mark CA-A as uncertain.

Table 17. Misclassified items on the worst-transferring shifts (XLM-R, wrong for all three seeds, correct for the target's own model), with English glosses by the authors. Drawn at random (seed 0) from all items of at most 22 words that meet these conditions; the complete candidate list is released.

Shift	Item	Gloss	Gold → pred.
HS-A→HS-C	حالیہ دنوں میں اس رہنما کے بیانات میں تضاد واضح ہے	Lately, this leader's statements have been clearly contradictory.	Hate → Normal
HS-A→HS-C	اکثر آن لائن ماحول میں برداشت کی کمی نظر آتی ہے	Online spaces often show a lack of tolerance.	Hate → Normal
FN-A→FN-B	دہلی میں این آر سی مخالف مظاہرے کے طور پر سلیمانی کا جنازہ نکلا۔	Soleimani's funeral procession shown as an anti-NRC protest in Delhi.	Fake → Real
FN-A→FN-B	ہندوستان کا وائرل نقشہ جموں و کشمیر کی غلط تقسیم کو ظاہر کرتا ہے۔	India's viral map shows a wrong division of Jammu and Kashmir.	Fake → Real
SA-C→SA-A	*bhut bekar delivery ha*	The delivery is very bad.	Negative → Positive
CA-B→CA-A	*tera kontent bakwas hy bhai*	Your content is rubbish, brother.	Abusive → Normal
CA-B→CA-A	*Aray mujhe ye vidiyo pasand aaya :)*	Oh, I liked this video :)	Normal → Abusive
PR-B→TW-D	*Honestly app bar bar crash karti hai update ke baad bhi*	Honestly, the app keeps crashing even after the update.	Negative → Positive

8 LIMITATIONS

(1) **Generated data.** 11 domains (9.7% of the benchmark) are LLM-generated, concentrated in hate speech, QA, cyber abuse and 3-class polarity. Cluster-aware splitting removes template leakage, but generated text carries a style signature that affects transfer: the generated opinion domains transfer well among themselves, in cyber abuse the label is partly predictable from length, and the shortcut audit shows that in HS-C the verb ہے ("is") occurs in 97.3% of normal items against 44.7% of hate items, a trace of how the two classes were generated. The organic-only analysis bounds these effects but cannot be run for Nastaliq sentiment or Roman cyber abuse. Generation prompts and the number of removed items were not recorded, and the full set of generated items was checked by one person; agreement with a second annotator is measured on a sample of 327 items only ($\kappa = 0.95$), and naturalness was not judged by a second person.

(2) **Provenance.** Three organic domains come from public uploads whose original authors could not be identified, so their annotation quality and licences are unverified.

(3) **Fake news.** Only two fake-news domains meet the inclusion criteria, so this matrix has two shifts and its conclusions, including the absence of useful transfer, rest on one pair of corpora.

(4) **Imbalanced domains.** Roman PR-B has 8 negative test items and fails the per-class minimum, which was added to the inclusion criteria after the experiments. It is kept for reproducibility, so scores involving it are noisy; Table 10 reports its matrix with and without it.

(5) **Models and prompts.** We test 7–8B LLMs in 4-bit precision with one prompt template, and encoders with one fixed set of hyperparameters; larger models, other prompts and tuned encoders may behave differently. Public organic corpora may also have been part of the LLMs' pretraining data, which would favour the LLMs on those domains.
(6) **LLM evaluation sample.** LLMs are scored on up to 300 items per target domain; the bootstrap intervals quantify this uncertainty for the family comparison, but differences within a family are not tested.
(7) **Error analysis.** The error categories are computed from the data, and the qualitative reading rests on a small random sample of examples rather than on a manual annotation of many errors.
(8) **Two tracks, not a script comparison.** The scripts differ in sources and task coverage, so the tracks are not directly comparable.

9 ETHICS STATEMENT

The benchmark contains offensive, hateful and sectarian text collected from public online sources, released to support moderation research; releases carry content warnings. Automated hate-speech and fake-news classifiers can suppress legitimate speech, especially that of minorities, and should not be deployed without human review. Sectarian and ethnic hate in the Pakistani context is particularly sensitive, and models trained on these data could also be misused to probe moderation systems. Our observation that sampled items of a generated hate-speech domain label mild criticism of public figures as hateful is a further reason not to deploy a classifier trained on them. No crowd workers were employed. All generated text, including generated hateful and abusive items, was read and checked by the first author. The second annotator, a fellow student, took part voluntarily after being told that the sample contains offensive and hateful text and that he could stop at any time. Repeated exposure to such material is a burden that future extensions with more annotators should plan for.

10 DATA STATEMENT

Following Bender and Friedman [7] and Gebru et al. [22]: **Language varieties:** Urdu in Perso-Arabic script (`urd-Arab`) and Roman Urdu (`urd-Latn`), mostly Pakistani usage, with frequent English code-mixing in the Roman track and some Punjabi in Nastaliq tweets. **Speakers:** authors of public tweets, YouTube comments, product and restaurant reviews, and news articles; generated text in 11 domains. **Annotation:** original corpus labels for organic data; labels for generated data were assigned in the generation prompt and checked by the first author, a native Urdu speaker; a second native speaker labelled a sample of 327 generated items blind ($\kappa = 0.95$). **Distribution:** GitHub (https://github.com/AbdullahPatti/Urdu-Multi-Domain-Script-Research), Zenodo (DOI 10.5281/zenodo.23053495) and Hugging Face.

11 CONCLUSION

In conclusion, Urdu classifiers are far less robust to a change of data source than their in-domain scores suggest, and the English finding of Calderon et al. [13] carries over to both Urdu scripts. Fine-tuned XLM-R and mBERT are the stronger in-domain classifiers in 6 of 7 matrices but lose 13.5–47.6 pp of macro-F1 on another domain, and in most shifts they do worse than a model trained on the target, so the loss is a robustness failure rather than a move to a harder domain. Few-shot LLMs lose much less and are significantly stronger across domains in 30 of 42 comparisons. The exceptions are informative: fine-tuning transfers better in QA validation, neither family transfers usefully in fake news across the two corpora available, and the largest failures of both families

occur where two domains attach the same label to different kinds of text, a problem that robustness to vocabulary or spelling would not solve. Reaching these conclusions required choosing domains by explicit criteria. Duplicated domains, labels that follow the origin of an item, question-level overlap and template near-duplicates each change cross-domain scores, and we recommend the audits released with this benchmark to anyone combining public and generated corpora into a multi-domain evaluation. None of these audits is specific to Urdu, so the construction procedure is likely to be as useful for other low-resource languages as the benchmark is for Urdu.

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A EVIDENCE OF LLM GENERATION IN EACH DOMAIN

This appendix supports the origin labels in Table 2. For every candidate domain it reports three signals that separate LLM-generated from human-written text in our data: templated near-duplicates, lexical diversity and the variability of item length (Section 3.2).

Table 18. Generation signals for all 33 candidate domains (train and test pooled). Counts (N) are for the candidate pool before the removals described in Sections 3.3 and 3.4, so they exceed the totals in Table 2. Template %: items with a near-duplicate variant (character 5-gram Jaccard ≥ 0.8) inside the domain. Distinct-2: distinct word bigrams per bigram in a 500-item sample. Len. CV: coefficient of variation of item length in words. Seed %: items near-duplicating an item of another domain of the same script. QA items are question–answer pairs, whose shared questions raise Template % by construction.

Domain	Name	N	Template %	Distinct-2	Len. CV	Seed %	Origin
Nas SA-A	Twitter Social Media	979	1.6	0.89	0.41	100.0	organic
Nas SA-B	Movie Reviews	9,163	0.3	0.47	0.82	8.9	translated
Nas SA-C	Civic Social Topics	162	0.0	0.73	0.15	0.0	generated, excluded
Nas SA-D	Governance Political	978	1.4	0.88	0.41	100.0	organic, excluded
Nas SA-E	Socioeconomic Agricultural	2,382	59.6	0.21	0.15	0.0	generated
Nas HS-A	Social Media Offensive	21,631	4.8	0.85	0.60	100.0	organic
Nas HS-B	Social Media Hate Speech	24,927	27.4	0.85	0.67	0.1	organic
Nas HS-C	Politics Sports Health Family	1,700	27.7	0.18	0.15	1.3	generated
Nas HS-D	Sports Labor Arts Education	1,700	41.2	0.24	0.17	0.3	generated
Nas HS-E	Inter Faith Sectarian Ethnic	21,629	4.8	0.83	0.60	100.0	organic, excluded
Nas FN-A	Multi Topic News	899	2.9	0.45	0.75	37.8	organic
Nas FN-B	Ax to Grind	9,812	13.0	0.65	1.98	4.5	organic
Nas FN-C	General Pakistani News	23,171	0.3	0.79	0.30	0.2	organic, excluded
Nas FN-D	Fact Checking Platform	13,388	0.4	0.55	0.46	0.0	organic, excluded
Nas QA-A	Religious Hadith	1,580	50.8	0.29	0.52	0.0	organic
Nas QA-B	General Knowledge	11,306	21.3	0.44	0.33	0.0	organic
Nas QA-C	Long Form QnA	1,120	51.8	0.10	0.09	0.0	generated
Nas QA-D	Short Form QnA	3,460	23.2	0.29	0.17	0.0	generated
Rom CA-A	YouTube Comments	4,868	25.9	0.56	0.29	6.4	uncertain
Rom CA-B	Content Creator Comments	1,300	67.5	0.29	0.18	0.0	generated
Rom CA-C	Social Media Diverse Vocab	1,300	63.2	0.32	0.17	0.0	generated
Rom CA-D	General Online Video	1,200	53.3	0.23	0.22	19.6	generated
Rom PR-A	Daraz Ecommerce	16,944	2.9	0.82	0.92	0.5	organic
Rom PR-B	Restaurant Food	750	0.5	0.74	0.65	0.4	organic
Rom PR-D	Electronics Gaming Delivery	26,666	23.2	0.93	1.10	2.8	organic
Rom SA-A	Mixed Public Discourse	21,444	0.5	0.95	0.98	2.8	organic
Rom SA-B	Social Media Politics Drama	10,993	1.6	0.89	1.18	100.0	organic
Rom SA-C	YouTube Entertainment	20,070	6.0	0.94	0.98	1.1	organic
Rom SA-D	Utilities Apps Courses	10,993	1.6	0.89	1.18	100.0	organic, excluded
Rom TW-A	Twitter Political Social	17,884	1.0	0.91	1.27	61.8	organic, excluded
Rom TW-B	Consumer Electronics Gaming	1,450	93.2	0.18	0.15	0.0	generated
Rom TW-C	Public Civic Services	1,450	91.9	0.19	0.13	0.0	generated
Rom TW-D	Mixed Social Commentary	1,600	92.4	0.23	0.12	0.0	generated

Table 19. Second-annotator check of the generated domains (Section 3.2): a stratified sample of 30 test items per domain, labelled without access to the original labels. Agreement and Cohen's κ are computed against the original labels; three items left blank by the annotator are excluded.

Domain	N	Agreement %	κ
Nas HS-C	30	96.7	0.93
Nas HS-D	30	100.0	1.00
Nas QA-C	30	100.0	1.00
Nas QA-D	30	100.0	1.00
Nas SA-E	30	100.0	1.00
Rom CA-B	30	90.0	0.80
Rom CA-C	28	100.0	1.00
Rom CA-D	29	93.1	0.86
Rom TW-B	30	90.0	0.85
Rom TW-C	30	93.3	0.90
Rom TW-D	30	90.0	0.85
All	327	95.7	0.95

B DOES A SINGLE PHRASE REVEAL THE LABEL?

This appendix supports the second inclusion criterion (Section 3.3). For each included domain it shows the phrase whose frequency differs most between classes. A content word is expected; a dateline, byline or other source marker would mean that the label can be read off the origin of an item.

Table 20. For every included domain, the word n-gram ($n \leq 3$, at least 20 occurrences) with the largest difference in document frequency between one class and the others. Content words are expected; source boilerplate (datelines, bylines) would indicate that the label follows the origin of an item.

Matrix	Domain	Phrase	Class	% of class	% of others
Sentiment (N)	SA-A	پر	Negative	17.1	9.0
Sentiment (N)	SA-B	خراب	Negative	26.4	7.5
Sentiment (N)	SA-E	دیا	Negative	45.3	4.7
Hate Speech (N)	HS-A	کو	Hate	38.9	30.1
Hate Speech (N)	HS-B	میں	Normal	34.8	24.8
Hate Speech (N)	HS-C	ہے	Normal	97.3	44.7
Hate Speech (N)	HS-D	صرف	Hate	28.0	0.0
Fake News (N)	FN-A	ہے۔	Real	41.8	25.3
Fake News (N)	FN-B	نہیں	Fake	32.6	9.8
QA Validation (N)	QA-A	وہ	Invalid	39.8	33.3
QA Validation (N)	QA-B	ہے	Valid	52.0	50.9
QA Validation (N)	QA-C	پیدا	Valid	12.9	9.1
QA Validation (N)	QA-D	ہے	Valid	30.3	27.7
Sentiment (binary) (R)	SA-A	*ha*	Positive	19.8	12.9
Sentiment (binary) (R)	SA-B	*ko*	Negative	31.8	17.6
Sentiment (binary) (R)	SA-C	*baap*	Negative	16.9	0.1
Cyber Abuse (R)	CA-A	*hai*	Abusive	22.1	10.7
Cyber Abuse (R)	CA-B	*nahi*	Abusive	27.8	3.1
Cyber Abuse (R)	CA-C	*nahi*	Abusive	30.1	0.0
Cyber Abuse (R)	CA-D	*hai*	Abusive	72.6	33.3
3-class Polarity (R)	PR-A	*but*	Neutral	53.3	6.7
3-class Polarity (R)	PR-B	*quality*	Negative	35.7	11.5
3-class Polarity (R)	PR-D	*allah*	Positive	12.9	2.7
3-class Polarity (R)	TW-B	*raha*	Negative	27.3	4.1
3-class Polarity (R)	TW-C	*nahi*	Negative	55.7	2.3
3-class Polarity (R)	TW-D	*nahi*	Negative	52.3	8.5

C SOURCE, URL AND LICENCE OF EACH DOMAIN

This appendix lists where each domain comes from, how its origin was confirmed, and the licence under which it is distributed (Section 3.5).

Table 21. Source and licence of every domain. Verified: our files were matched against the public release (content or file identity). Untraced: obtained from a public upload whose original author could not be identified. Generated: produced by the first author with xAI Grok (Section 3.2).

Domain	Origin	Source	Description	URL	Licence
Nas SA-A	organic	verified	Urdu Sentiment Corpus v1.0 (Khan & Nizami 2020) [32]	https://github.com/MuhammadYaseenKhan/Urdu-Sentiment-Corpus	none stated (citation requested)
Nas SA-B	translated	verified	IMDB reviews machine-translated into Urdu (Hugging Face upload by mirfan899); 91% of items match after normalising punctuation and letter variants [42]	https://huggingface.co/datasets/mirfan899/imdb_urdu_reviews	ODbL 1.0
Nas SA-E	generated	generated	Grok-generated by first author	–	xAI terms (attribute Grok)
Nas HS-A	organic	verified	ISE-Hate corpus (Akram, Shahzad & Bashir 2023); same 21,361 tweets, 98.7% label agreement with Level-1 [2]	https://github.com/hammad7007/ISE_dataset	none stated
Nas HS-B	organic	verified	Urdu hate-speech tweets (Hugging Face upload by Adnan855570); our raw file is byte-identical [1]	https://huggingface.co/datasets/Adnan855570/Urdu_Hate_Speech	MIT
Nas HS-C	generated	generated	Grok-generated by first author	–	xAI terms (attribute Grok)
Nas HS-D	generated	generated	Grok-generated by first author	–	xAI terms (attribute Grok)
Nas FN-A	organic	verified	Bend the Truth (Amjad et al. 2020) [3]	https://github.com/MaazAmjad/Datasets-for-Urdu-news	none stated
Nas FN-B	organic	verified	Ax-to-Grind Urdu (Harris et al. 2023) [26]	https://github.com/HjH-Whu-CRC/Ax-to-Grind-Urdu	none stated
Nas QA-A	organic	verified	mirfan899/Urdu GitHub collection (MIT); 100% content match [41]	https://github.com/mirfan899/Urdu	MIT
Nas QA-B	organic	verified	mirfan899/Urdu GitHub collection (MIT); 100% content match [41]	https://github.com/mirfan899/Urdu	MIT
Nas QA-C	generated	generated	Grok-generated by first author (70 questions x 8 variants)	–	xAI terms (attribute Grok)
Nas QA-D	generated	generated	Grok-generated by first author	–	xAI terms (attribute Grok)
Rom SA-A	organic	verified	HF ayeshasameer/roman-urdu-sentiment-analysis (MIT); identical test split [4]	https://huggingface.co/datasets/ayeshasameer/roman-urdu-sentiment-analysis	MIT
Rom SA-B	organic	verified	RUSAD (Mehmood Essam & Shafi 2019) [38]	https://archive.ics.uci.edu/dataset/854/roman+urdu+sentiment+analysis+dataset+(rusad)	CC BY 4.0
Rom SA-C	organic	untraced	YouTube comments; uploader unknown	–	unknown
Rom CA-A	uncertain	untraced	Unknown; shows augmentation noise	–	unknown
Rom CA-B	generated	generated	Grok-generated by first author	–	xAI terms (attribute Grok)
Rom CA-C	generated	generated	Grok-generated by first author	–	xAI terms (attribute Grok)
Rom CA-D	generated	generated	Grok-generated by first author (seeds partly from CA-A: 20% near-matches)	–	xAI terms (attribute Grok)
Rom PR-A	organic	verified	Daraz Code-Mixed Product Reviews (Kaggle) [53]	https://www.kaggle.com/datasets/yrrebeere/daraz-code-mixed-product-reviews	CC BY 4.0
Rom PR-B	organic	untraced	Kababjees restaurant reviews	–	unknown
Rom PR-D	organic	verified	RUWV-NSR, Soomro et al. 2024 Data in Brief; identical file name and byte size [49]	https://data.mendeley.com/datasets/v5jfhsvtmd/4	CC BY 4.0
Rom TW-B	generated	generated	Grok-generated by first author	–	xAI terms (attribute Grok)
Rom TW-C	generated	generated	Grok-generated by first author	–	xAI terms (attribute Grok)
Rom TW-D	generated	generated	Grok-generated by first author	–	xAI terms (attribute Grok)

D PROMPT USED FOR THE FEW-SHOT LLMS

The user message sent to every LLM, before the model's chat template is applied. <INSTRUCTION> is "Classify the sentiment of the text.", "Decide whether the text is abusive or hateful.", "Decide whether the news text is real or fake." or "Decide whether the answer is a valid answer to the question."; <SCRIPT> is "Nastaliq (Perso-Arabic) script" or "Roman (Latin) script"; <LABELS> lists the label words of the matrix (e.g. "Negative or Positive", "Negative, Neutral or Positive"). Each demonstration and the test item are rendered as `Text: …`, or as `Question: …` and `Answer: …` lines for QA, followed by `Label:`.

```
<INSTRUCTION> The text is Urdu written in <SCRIPT>. Answer with exactly
one word: <LABELS>.

Examples:

Text: <demonstration 1>
Label: <label 1>

... (5 demonstrations per class, shuffled)

Now classify:

Text: <test item>
Label:
```